

Chapter 3

Word Problems I: Building Mathematical Models

Opening Challenge

A student has scores of 76, 83, and 88 on three quizzes. Two quizzes remain, and they will have equal weight. What average is needed on the last two quizzes to finish with an overall average of 84?

Before calculating, ask a more important question: *What should the unknown represent?*

If x is the average on the last two quizzes, then their total score is $2x$. Thus

$$\frac{76 + 83 + 88 + 2x}{5} = 84.$$

Solving gives

$$247 + 2x = 420, \quad 2x = 173, \quad x = 86.5.$$

The algebra is easy once the model is correct. This is the central idea of the chapter: a word problem is not mainly a test of calculation. It is a test of whether you can translate a situation into mathematics.

3.1 From Words to Mathematics

A useful modeling process has five steps.

1. **Understand the question.** What is known? What must be found?
2. **Name the unknown.** Choose a variable and include its unit.
3. **Build a relationship.** Translate the important sentence into an equation.
4. **Solve the equation.**

5. **Interpret and check.** Does the answer make sense in the original situation?

Many mistakes happen before any algebra begins. A student may choose an unclear variable, compare the wrong quantities, forget a unit, or apply a percent to the wrong base. Good problem solving therefore begins with reading and organizing, not with manipulating symbols.

Professor Shen's Remark. After solving a problem, do not stop. Ask: "What was the key relationship?" "Could I change the numbers and solve a whole family of problems?" "Can I derive a formula?" These questions turn an exercise into mathematical training.

3.2 Averages and Totals

The average formula

$$\text{average} = \frac{\text{total}}{\text{number of items}}$$

is often more useful when rewritten as

$$\text{total} = (\text{average})(\text{number of items}).$$

Example 3.1. Four measurements have average 18. A fifth measurement is added, and the new average becomes 20. Find the fifth measurement.

Solution. The first four measurements have total

$$4 \cdot 18 = 72.$$

Let x be the fifth measurement. The new total is $5 \cdot 20 = 100$, so

$$72 + x = 100.$$

Hence

$$x = 28.$$

Discovery. Suppose n numbers have average A . One new number is added and the average becomes B . If the new number is x , then

$$nA + x = (n + 1)B,$$

so

$$\boxed{x = (n + 1)B - nA}.$$

We have derived a formula from one simple word problem.

3.3 Percent Problems: Always Identify the Base

The statement " $r\%$ of Q " means

$$\frac{r}{100}Q.$$

If a quantity increases by $r\%$, its new value is

$$\left(1 + \frac{r}{100}\right) Q.$$

If it decreases by $r\%$, its new value is

$$\left(1 - \frac{r}{100}\right) Q.$$

The most important question is: *percent of what?*

Example 3.2. A jacket is marked down 20%, and the reduced price is then marked down another 10%. The final price is \$144. Find the original price.

Solution. Let x be the original price. After the first discount the price is $0.80x$. After the second discount it is

$$0.90(0.80x) = 0.72x.$$

Thus

$$0.72x = 144,$$

so

$$x = 200.$$

Notice that the total discount is not 30%. The second 10% is taken from the already reduced price.

Successive Percent Changes

If a quantity is multiplied successively by factors a and b , the total effect is multiplication by ab . For example, an increase of 20% followed by a decrease of 20% gives

$$1.20 \cdot 0.80 = 0.96.$$

The final amount is 96% of the original, not 100%.

Ask a Question. If a price rises by $p\%$, what percent decrease is needed to return to the original price? Try to derive the answer before looking for a numerical example.

3.4 Mixtures and Conservation

Mixture problems become much easier when we track the amount of the ingredient that matters.

$$\text{amount of ingredient} = (\text{concentration})(\text{total amount}).$$

The basic principle is conservation:

$$\text{ingredient before} + \text{ingredient added} = \text{ingredient after}.$$

Example 3.3. A container holds 12 liters of a 25% salt solution. How many liters of pure water should be added to make the solution 20% salt?

Solution. Initially the amount of salt is

$$0.25 \cdot 12 = 3 \text{ liters.}$$

Let x liters of water be added. The amount of salt stays 3 liters, while the total volume becomes $12 + x$. Therefore

$$\frac{3}{12 + x} = 0.20.$$

Thus

$$3 = 2.4 + 0.2x, \quad x = 3.$$

So 3 liters of water should be added.

Example 3.4. A tank contains 20 liters of a 30% antifreeze mixture. How many liters of the mixture should be removed and replaced by pure antifreeze to obtain a 40% mixture?

Solution. Let x liters be removed. Since the removed liquid is 30% antifreeze, the tank loses $0.30x$ liters of antifreeze. Replacing it with x liters of pure antifreeze adds x liters of antifreeze. The final amount of antifreeze is therefore

$$6 - 0.30x + x = 6 + 0.70x.$$

The total volume remains 20 liters. Hence

$$\frac{6 + 0.70x}{20} = 0.40.$$

So

$$6 + 0.70x = 8, \quad x = \frac{20}{7}.$$

Thus $20/7$ liters, about 2.86 liters, must be replaced.

Derive a General Formula

Suppose a container holds V liters of a mixture with concentration a , where a is written as a decimal. We remove x liters and replace it with a pure ingredient. If the desired concentration is b , then

$$a(V - x) + x = bV.$$

Solving,

$$aV - ax + x = bV,$$

$$(1 - a)x = (b - a)V,$$

and therefore

$$\boxed{x = \frac{(b - a)V}{1 - a}}.$$

A formula is not magic. It is a compressed record of reasoning.

3.5 Age Problems and Time Shifts

Age problems are really problems about adding or subtracting the same amount of time from people's present ages.

Example 3.5. Six years from now, Maya will be twice as old as she was four years ago. How old is she now?

Solution. Let x be Maya's present age. Six years from now she will be $x + 6$, and four years ago she was $x - 4$. Thus

$$x + 6 = 2(x - 4).$$

Solving gives

$$x + 6 = 2x - 8, \quad x = 14.$$

A check is immediate: in six years she will be 20, and four years ago she was 10.

3.6 Ratios and Parts

A ratio such as

$$5 : 3 : 2$$

means that the three quantities may be written as

$$5t, \quad 3t, \quad 2t$$

for some common scale factor t .

Example 3.6. Red, blue, and green beads are mixed in the ratio 4 : 3 : 5. There are 72 beads altogether. How many are green?

Solution. Write the numbers as $4t$, $3t$, and $5t$. Then

$$4t + 3t + 5t = 72,$$

so

$$12t = 72, \quad t = 6.$$

The number of green beads is

$$5t = 30.$$

Generalization. If quantities are in the ratio $a : b : c$ and their total is T , then the first quantity is

$$\frac{a}{a + b + c}T,$$

and similarly for the other two. Derive this formula yourself from $at + bt + ct = T$.

3.7 Distance, Rate, and Relative Motion

The basic relation is

$$\boxed{d = rt},$$

where d is distance, r is rate, and t is time.

Always make the units consistent before using the formula.

Example 3.7. A cyclist travels at 12 miles per hour. A second cyclist starts from the same point 30 minutes later and travels along the same road at 16 miles per hour. How long after the second cyclist starts will the second cyclist catch the first?

Solution. The first cyclist has a head start of

$$12 \left(\frac{1}{2} \right) = 6 \text{ miles.}$$

The second cyclist gains at the relative rate

$$16 - 12 = 4 \text{ miles per hour.}$$

If t is the number of hours after the second cyclist starts, then

$$4t = 6.$$

Thus

$$t = \frac{3}{2}.$$

The second cyclist catches the first after 1.5 hours.

Discovery. If one traveler has a head start of D miles and the faster traveler gains at v miles per hour, the catch-up time is

$$\boxed{t = \frac{D}{v}}.$$

Here v is a *relative speed*, not either traveler's actual speed.

3.8 How to Check a Mathematical Model

An algebraic answer can be correct while the model is wrong. Use the following checks.

1. **Units:** Are dollars compared with dollars, hours with hours, and liters with liters?
2. **Size:** Is the answer reasonably large or small?
3. **Sign:** Can the quantity be negative in the original problem?
4. **Substitution:** Does the answer satisfy the relationship described in words?
5. **Alternative reasoning:** Can you estimate or solve the problem in a second way?

3.9 How Mathematicians Learn from a Solved Problem

A strong student should be able to do four things after solving a problem.

1. **Summarize:** State the main idea in one or two sentences.
2. **Generalize:** Replace the numbers by variables and derive a formula.
3. **Vary:** Change one condition and ask what happens.
4. **Create:** Write a new problem that uses the same mathematical idea but has a different story and structure.

For example, after solving a mixture problem, ask what changes if pure water is added instead of pure acid. After solving a catch-up problem, ask what happens if the two travelers move toward each other. After solving an average problem, ask how one new observation changes the old average.

These habits are part of the training of a future scientist or mathematician. Science advances not only by answering questions, but also by learning which questions are worth asking.

What Have We Discovered?

1. Word problems are mathematical modeling problems: choose quantities, identify relationships, and translate them into equations.
2. The total corresponding to an average is

$$(\text{average})(\text{number of items}).$$
3. Percent changes are multiplication factors, and successive percent changes multiply rather than add.
4. Mixture problems are governed by conservation of the ingredient being tracked.
5. Ratios can be represented using a common scale factor.
6. Distance problems are organized by $d = rt$, and catch-up problems often become simple after using relative speed.
7. A solution should be checked against units, size, signs, and the original story.
8. The deepest step is often after the solution: summarize, generalize, change assumptions, and create a new question.

Do not only solve the model. Learn how to build, test, and improve it.

Exercises

Problems 1–10 are Core Exercises. Problems 11–20 are Challenge Problems at approximately AMC 8 and MATHCOUNTS level. All twenty problems have been newly written for this book rather than copied from previous contests.

Part A. Core Exercises

1. Five numbers have average 24. Four of the numbers are 17, 21, 26, and 31. Find the fifth number.
2. A student has test scores 82, 87, and 91. What score is needed on a fourth test to make the average exactly 88?
3. A backpack originally costs \$60. Its price is reduced by 15%. Find the sale price.
4. After a 20% increase, a monthly fee is \$54. What was the fee before the increase?
5. A 10-liter solution is 30% juice. How many liters of pure water must be added to make the mixture 20% juice?
6. A container has 15 liters of a 40% salt solution. How many liters of pure salt solution must be added to make the mixture 50% salt?
7. Seven years from now, Leo will be three times as old as he was five years ago. Find Leo's present age.
8. The numbers of three kinds of books are in the ratio 2 : 3 : 7. There are 96 books altogether. How many books are in the largest group?
9. A runner travels 6 miles at a constant speed of 8 miles per hour. How many minutes does the trip take?
10. A car travels 150 miles. It travels the first 60 miles at 40 miles per hour and the remaining 90 miles at 60 miles per hour. How long does the entire trip take?

Part B. Challenge Problems

11. The average of eight numbers is 15. When the largest number is removed, the average of the remaining seven numbers is 12. What number was removed?
12. A store increases the price of an item by 25%. By what percent of the new price must the store reduce it to return exactly to the original price?
13. A tank contains 24 liters of a 25% solution. Some of the mixture is removed and replaced by pure concentrate. The final mixture is 40% concentrate. How many liters were replaced?
14. Nora is four years older than Eli. Six years ago, Nora was twice Eli's age. Find their present ages.

15. A box contains red, white, and blue tiles in the ratio 3 : 4 : 5. After 6 red tiles are added, the ratio of red tiles to the total number of tiles is 1 : 3. How many tiles were originally in the box?
16. A bus leaves a station at 9:00 A.M. traveling at 45 miles per hour. A car leaves the same station at 9:40 A.M. along the same road at 60 miles per hour. At what time does the car catch the bus?
17. A 120-point contest has two kinds of problems. Each regular problem is worth 4 points and each challenge problem is worth 7 points. There are 24 problems altogether. If all problems are answered correctly, how many challenge problems are there?
18. A restaurant bill consists of the meal price, an 8% tax on the meal, and then a 20% tip calculated on the meal plus tax. The final amount is \$64.80. What was the price of the meal before tax and tip?
19. A ball rebounds to $\frac{2}{3}$ of the height from which it falls. Its third rebound reaches a height of 16 inches. From what height, in inches, was the ball originally dropped?
20. A class has n students with average score A . One additional student with score s joins the class, and the new average is B .
 - (a) Derive a formula for s in terms of n , A , and B .
 - (b) Use your formula when $n = 19$, $A = 78$, and $B = 79$.
 - (c) Write one new question about averages that can be answered using the same idea.

Beyond the Chapter

These questions are for exploration rather than speed.

1. A quantity is increased by $p\%$. Derive a formula for the percent decrease needed to return to the original value. Test your formula with $p = 25$.
2. A container holds V liters of a mixture with concentration a . You add x liters of pure water and want concentration b , where $b < a$. Derive a formula for x .
3. Two travelers begin D miles apart and move toward each other at speeds u and v . Derive a formula for the meeting time. Explain why the relevant relative speed is $u + v$ rather than $|u - v|$.
4. Choose one problem from this chapter. Write a three-sentence summary: (1) what quantity you chose as the unknown, (2) what relationship produced the equation, and (3) what general lesson the problem teaches. Then change one condition and create a new problem of your own.

Answer Key

1. 25.
2. 92.
3. \$51.
4. \$45.
5. 5 liters.
6. 3 liters.
7. 11 years old.
8. 56 books.
9. 45 minutes.
10. 3 hours.
11. 36.
12. 20%.
13. 4.8 liters.
14. Eli is 10 and Nora is 14.
15. 48 tiles.
16. 11:40 A.M.
17. 8 challenge problems.
18. \$50.
19. 54 inches.
20. (a) $s = (n + 1)B - nA$. (b) $s = 98$. (c) Answers will vary.

Notes on the Exploration Problems

For the first exploration problem, an increase by $p\%$ multiplies the original amount by $1 + p/100$. If the required decrease is $q\%$, then

$$\left(1 + \frac{p}{100}\right) \left(1 - \frac{q}{100}\right) = 1.$$

Solving gives

$$q = \frac{100p}{100 + p}.$$

For $p = 25$, this gives $q = 20$.

For the second exploration problem, conservation of the ingredient gives

$$aV = b(V + x),$$

so

$$x = \frac{(a - b)V}{b}.$$

For the third exploration problem, the distance between the travelers shrinks at $u + v$ miles per hour, so

$$t = \frac{D}{u + v}.$$